# Transistors, relays & MOSFETs

ENGINEERING WITH MR KELLY

Higher · Electronics & control · Worksheet

Name _____ Class _____ Date _____

**By the end of this worksheet I can…**

- ☐ describe how a transistor works as a switch (about 0.7 V base–emitter)
- ☐ explain the purpose of a relay and a protection diode
- ☐ calculate base current, collector current and current gain (hFE)
- ☐ describe MOSFET threshold voltage and an advantage of a MOSFET over a BJT
- ☐ use a MOSFET graph to find drain–source current

## I DO — Worked example

Read how a full-mark answer is built

*A BJT has a 2.2 kΩ base resistor. The input voltage is 5 V and Vbe = 0.7 V. hFE = 150.*

**1 Calculate the base current and collector current.** (3 marks)

**Full-mark answer** each ✓ is one mark

VRb = 5 − 0.7 = 4.3 V ✓. Ib = 4.3 ÷ 2200 = $1.95 \times 10^{-3}$ A ✓. Ic = hFE × Ib = 150 × $1.95 \times 10^{-3}$ = 0.29 A ✓.

**Why marks are not awarded**

*"Ib = 5 ÷ 2200 = 2.27 mA, Ic = 150 × 2.27 mA = 0.34 A"*

1 mark (follow-on for Ic): the 0.7 V drop was missed.

## WE DO — Guided practice

Use the starters and word bank to help you

*An engineer replaces a BJT with a MOSFET.*

**2 Describe one advantage of the MOSFET over the BJT.** (1 mark)

**Sentence starters**

One … is …, which means …

Another … is …, so …

**Word bank**

higher · almost · currents · little · current · switch · resistance · controller

Tip: 1 marks = 1 separate, developed points.

**YOU DO** **On your own** No notes. Check your answers afterwards.

### 3 Quick check: tick one box for each.

**a)** Why is a diode placed across a relay coil?
☐ To light up ☐ To protect the transistor from reverse voltage ☐ To increase current ☐ To switch the relay

**b)** A MOSFET is controlled by…
☐ Base current ☐ Gate voltage ☐ Collector current ☐ Light

**c)** IB = 2 mA and hFE = 150. IC is…
☐ 75 mA ☐ 300 mA ☐ 152 mA ☐ 0.3 mA

**d)** Which component lets a 5 V circuit switch a 230 V heater?
☐ LDR ☐ Relay ☐ Diode ☐ Buzzer

### 4 Key terms: write the letter of the matching description next to each term.

| | | |
|---|---|---|
| 1. **Vce** | ____ | **A** Vcc − VL (Kirchhoff's second law) |
| 2. **MOSFET saturation** | ____ | **B** Lets a low-current circuit switch a separate high-current circuit. |
| 3. **Transistor switch-on voltage** | ____ | **C** Limits base current to protect the transistor. |
| 4. **Base resistor** | ____ | **D** VDS = VGS − VT |
| 5. **Relay** | ____ | **E** About 0.7 V at the base. |
| 6. **Protection diode** | ____ | **F** Across the relay coil; protects the transistor from reverse voltage when the coil switches off. |

*A MOSFET has a threshold voltage of 2.1 V.*

### 5 Describe what is meant by threshold voltage. (1 mark)

*A transistor drives a relay coil. A diode is connected across the coil.*

### 6 Explain the purpose of the diode. (2 marks)

*A relay is used between a transistor and a mains lamp.*

**7 Explain why a relay is used.** (2 marks)

**Check and reflect**

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?